

Evaluating an Itinerary Recommendation Algorithm for Runners

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ABSTRACT

Recommender systems for runners primarily rely on existing running traces in an area. In the absence of running traces, recommending running routes is challenging. This paper describes our approach to generating and proposing "pleasant" running tours that consider the runner's standard preferences and their distance and elevation constraints. Our algorithm is an approach to solve the cold start recommendation problem in unknown places by mining available map-data. We implemented a prototypical smartphone app that generates and recommends pleasant running routes to evaluate our algorithm's effectiveness. An in-the-wild user study was conducted, with 11 participants across three cities. We tested the correlation between what is defined as "pleasant path" by our algorithm and the user's perception. The results of the user study show a positive correlation and support our algorithm. We also outline implications for the design of successful recommendation algorithms for runners.

CCS CONCEPTS

• **Information systems** → *Recommender systems*; • **Human-centered computing** → *User studies*; • **Theory of computation** → *Design and analysis of algorithms*.

KEYWORDS

Itinerary Recommendation, Running, Route Generation, Sports

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1 INTRODUCTION

In HCI, there is a significant increase in research to understand and design interactive technologies that support sports such as running. Running is one of the most popular forms of exercise around the world. It does not require special equipment or complex organization and can be practiced almost anywhere. However, it can be difficult for users to find running routes on their own without external assistance, especially in unknown environments. Runners mostly stick to a few usual itineraries instead of discovering and trying out new ones [8, 13]. The runner is often limited in exploring an area even with the available external assistance, as it is not easy to find a novel, pleasant itinerary that matches their constraints and targets in terms of distance and difficulty. Indeed, itinerary identification and recommendation is a complex problem.

Few available commercial systems [5, 7] propose new running itineraries by exploiting user-generated content, i.e. existing collections of running traces. The user can also search among the routes recorded by someone else in the platform's community or create their own route on a map. However, these have many limitations. Relevant running traces are not always available in a given area. Even in dense areas, people increasingly want to protect their privacy and prefer not to share their traces. In such situations, the recommendation systems face a cold-start problem. It becomes difficult or even impossible to construct a tour based on such limited data.

Our work aims to develop a tool to propose pleasant round-trip tours to runners in unknown locations, similar to [6, 9, 10, 15] considering the standard preferences of a runner. We created a solution that automatically generates novel pleasant itineraries by mining map data without requiring the availability of running traces or the user's personal history. Another objective was to encourage the exploration of new paths in the area. Our algorithm mines OSM (Open Street Map) data for route characteristics and gives weight to these characteristics to generate pleasant running route for runners. While there are a few approaches that generate itineraries using only map data [2, 3, 14], they remain quite basic, (e.g. generating itineraries based on finding the shortest path between a few points of interest). A few others take one [12] or many route characteristics into account [6] but they do not consider them for an

itinerary generation; they only use them to filter or rank the candidate itineraries selecting the best ones. Moreover, these works have not been validated by user studies except for [6], where feedback was recorded only on the overall quality of the tour. In contrast, our work conducts a more in-depth analysis of our algorithm by assessing the overall tour quality and the scores assigned by our system to a path segment according to different characteristics.

We conducted a user study in an uncontrolled environment to test our algorithm's effectiveness. We designed and developed a mobile application that generates pleasant tours using our algorithm. Each tour consists of segments (or paths) of different qualities, and the mean score of these segments defines the overall tour quality. We wanted to test our algorithm's scoring function, i.e., the weights/scores assigned to different characteristics of the path, and validate that it is positively correlated with the user's appreciation. Therefore, we asked participants to run three routes proposed by our system and give feedback on several pre-selected segments of interest in each route, by rating them on a five-star Likert scale. We then analyzed the correlation between the value of the given user rating and the score of the segments computed by our algorithm. In the following, we detail our algorithm, the development of the mobile prototype we used for the evaluation campaign, the user experiment we carried out, and our findings and insights for future work.

2 ITINERARY RECOMMENDATION ALGORITHM

To generate pleasant itineraries, we started by formalizing what is a pleasant path for a runner. We conducted user interviews with seven regular runners, where they listed the path characteristics they preferred for running. From these preliminary user interviews and internal discussions with the team members, almost all regular runners, we specified the runners' standard objectives (distance, elevation gain), path preferences (type, environment: closeness to nature vs traffic, maximum steepness) and overall tour shape for our algorithm.

For our algorithm, we consider the pleasantness, or quality, of an itinerary as the extent to which its characteristics match runners' preferences. To assess this, we analyze the quality of the path segments that compose the itinerary based on the specified characteristics, translated into a numerical score to be used as input for the generation algorithm. A post-processing step was included to improve the shape of the tour by avoiding backtracking and too many turns.

Using OSM data and standard routing libraries, we started by building a routing graph. The graph's nodes are the road crossings, and the arcs are the road segments that link the crossings. We computed a pleasantness score for each arc based on OSM attributes, according to standard runners' preferences. For example, paths along hiking trails, parks or riversides were assigned a much higher score than paths close to main roads or highways. In our prototype implementation, we mined three aspects from OSM maps and one aspect from IGN (Institut Geographic National) data:

- **Proximity to nature:** A score from 0 to 1, depending on the segment's length within or close to a nature area (e.g., park, forest, beach, lake or river border).

- **Type of path:** A score from 0 to 1, depending on the path type: tracks and paths are 1, pedestrian streets are 0.75, while secondary roads are 0.5.
- **Distance to high traffic roads:** A score from 0 to 1, depending on the path distance to high traffic roads (main road, motorway).
- **Path steepness:** We also assess how suitable each path segment's steepness (computed from IGN elevation information) is according to the elevation profile selected by the user (flat, hilly or alpine) and translate it into a penalty (to avoid steepness greater than 5%, 10% or 20% respectively)

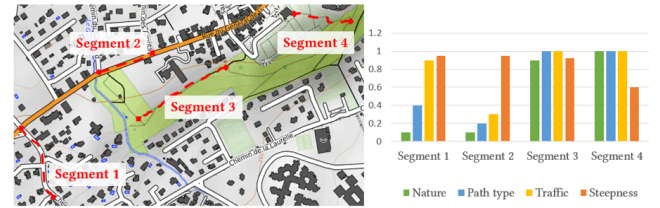


Figure 1: Example of scores affected to four segments of a path – for instance, Segment 1 has a low nature score, a below-average path type score but does not have any traffic or steepness issue.

Given the scored routing graph, a starting point, a target distance and elevation gain, we focused on generating appropriate tours, i.e. cycles of the requested length, that maximize the pleasantness score on the visited segments. This optimization problem can be formulated as an arc-orienteeing problem. It can be seen as a combination of two well-known NP-hard (non-deterministic polynomial time) combinatorial problems: (i) a knapsack problem, where the goal is to select on a map the best locations for the runner within a (distance) budget constraint; and (ii) a travelling salesman problem, where the goal is to find the shortest pleasant route that visits these locations.

3 EVALUATION

The usability evaluation of mobile recommender systems such as running tours recommenders should consider the real environment to test if the maps provide sufficient information and the specific context, i.e., "mobility". These aspects cannot be satisfactorily simulated in a laboratory. The evaluation of our algorithm requires the user understanding and perception of its environment. Hence, we decided to conduct an asynchronous remote experiment in a real (uncontrolled) environment. In the asynchronous user testing method, the researchers do not have access to the data in real-time, and no facilitator is interacting with the user during data collection. The qualitative data was collected through post-test self-report forms and emails. The asynchronous approach allowed us to experiment during difficult coronavirus COVID-19 pandemic [11] times while leveraging the real-world context.

Through our experiment, we wanted to evaluate the following hypotheses:

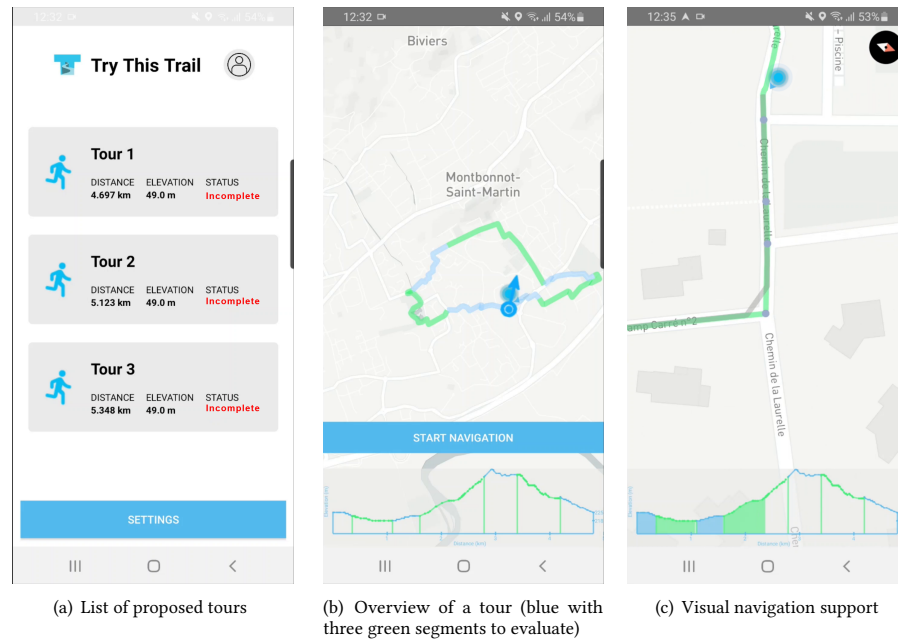


Figure 2: The mobile app interface

- (1) The segment score given by the algorithm and the user ratings have a positive correlation. The path characteristic having a higher score in the segment of interest is the reason for the user's segment preference and vice-versa.
- (2) The running tour recommendations enable and improve area exploration for the users.

3.1 Experiment design and implementation

3.1.1 Granularity of quality evaluation. In our experiment, we wanted to evaluate our algorithm's scoring functions. Our algorithm produces tours by combining segments of variable scores. Therefore, for our purpose, evaluating only of the overall tour score was not sufficient. We wanted to compare the score given by our system to specific path segments with the user's rating of these segments. Hence, during the experiment, we asked the users for feedback on manually selected segments with a given average length, and a stable and consistent score, i.e. a score that did not vary too much, over this segment. We manually selected these segments of interest to ensure that we would ask the user to rate a set of segments covering a range of different path characteristics and score values.

3.1.2 Prototype supporting the experiment. For our experiment, we developed an Android mobile prototype through which a user could request tours with a specific distance, elevation profile (flat, hilly, alpine) and starting point. This request was then sent to our backend where our algorithm generated ten "pleasant" tours. Out of the initially generated ten tours we first manually selected three tours to avoid tour overlap and cover different geographical areas. For each of these tours we then identified three segments of interest that had a stable score over 300-800 m but different path

characteristics and corresponding score values. We finally sent back these tours to the user, asking him or her to run them and to rate the selected segments of interest. The mobile app displayed the three tours and the status of each of them as complete or incomplete (Fig. 2(a)). Once selected, the running tour was displayed in blue with the three segments of interest shown in green (Fig 2(b)). Visual navigation support with haptic notification was provided in the app. The current location and the running tour were displayed on a map (Fig. 2(c)), and the user was informed when entering or exiting a segment of interest through vibration. As mentioned in the prior art, we acknowledge that visual navigation is not the best support for running [4]. However, this experiment aimed to test the correlation between algorithm and user scores rather than a complete running app.

3.1.3 Measures. When exiting, users were asked to rate the segment on a five-star scale. After this feedback, they continued their run. At the end of the tour, they rated the overall tour, ranked the segments according to their preference and answered a multiple-choice question about why they liked their preferred segment (e.g. because of nature, the path type etc.) and why they disliked the least rated segment. They could also give feedback through open comment boxes for additional reasons or any issues they encountered. The user data collected through the app in the form of app logs, ratings and comments were sent to our database for analysis.

3.1.4 Procedure. Twenty-four participants were recruited initially through snowball sampling in three cities in France: a large metropolitan city, a medium-sized mountainous city and a smaller city on the seaside. However, due to coronavirus confinement circumstances (delays and additional constraints), only 11 participants (5 male and 6 female, aged from 20 to 55 years) could participate

in the test. All participants ran more than once a week between 5-10kms. To compensate for their efforts, every participant received a voucher worth 25 EUR. The participants were briefed about the experiment protocol and terms of user participation via email. After agreeing to the terms about their data collection and use, a user manual with detailed instructions related to the app usage and the link to install the application was sent.

Every participant installed the mobile application on their android device as per instruction. Since we also wanted to test the quality of area exploration enabled by the algorithm, we asked our participants to select their starting point away from their home or usual running location to run in unfamiliar segments. Ten participants completed all the three running tours. However, one participant was only able to complete one tour as, during the other tours, the app had severe technical issues. All participants sent their ratings and feedback after finishing each tour, which we analyzed for correlation.

4 RESULTS

4.1 Quantitative

The rated segments (10 participants x 3 tours x 3 segments of interest + 1 participant x 1 tour x 3 segments of interest = 93 segments of interest) were analyzed for correlation to the score. We standardized (centring and scaling to unit variance) his/her ratings and the score of segments s/he ran for each user. To better normalize the ratings amongst participants, we initially thought of asking runners to run a common tour, but this ended up being too complicated to organize in our campaign. After a simple normalization, a moderate correlation of **0.46** was found between the user ratings and the mean segment score (Fig. 3). A more nuanced analysis showed that navigation experience problems or other issues described in the Limitations section below could explain some outlier ratings.

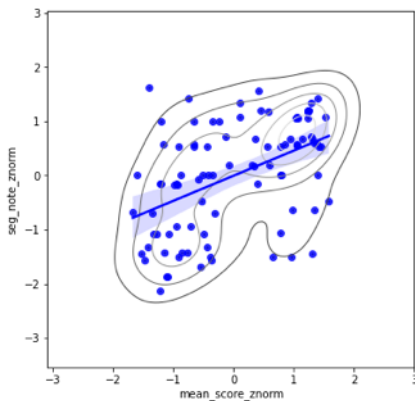


Figure 3: Regression and Contour Plot of the standardized mean score per segment against the use of standardized rating, which shows a correlation of 0.46 between the variables

An analysis of the three most influential score characteristics (nature, path type, traffic) that was each mentioned as decisive in more than 2/3 of the rated segments) has proven that their values are consistent with user's perception. If we analyze the answers to

the multiple-choice question about why they liked their preferred segment, we get the following (0=bad to 1=good):

- Segments that **users rated high** because of **nature** had an average nature score of **0.72**
- Segments that **users rated low** because of **nature** had an average nature score of **0.10**
- Segments that **users rated high** because of **path type** had an average path type score of **0.76**
- Segments that **users rated low** because of **path type** had an average path type score of **0.40**
- Segments that **users rated high** because of **traffic** had an average traffic score of **0.99**
- Segments that **users rated low** because of traffic had an average **traffic** score of **0.89**

We note that a very high traffic score was observed in the generated tours, hinting either that the high traffic roads were rare in the area where the tour was generated, or our algorithm penalized the proximity to such roads strongly.

The moderate correlation and analysis of the score characteristics provide positive validation for the first hypothesis.

4.2 Qualitative

The qualitative data was collected through open comment boxes in the mobile application and email feedback. Overall, the participants (7/11) appreciate the concept and quality of tour recommendation. The tours recommended to users had running paths that were unknown to them. 5/11 participants appreciated discovering new places which provides partial support to our second hypothesis. A few raised concerns on the recommendations as the (2/11) participants ran a tour which went through areas unsafe to be run at night. Although safety was indeed mentioned as an important criterion during the initial interviews, no reliable data sources were available to integrate it into our algorithm. A few participants also did not like the path segments that were near hospitals (1/11), a train station (4/11) and a large car park (1/11) because of their negative ambience. One participant preferred explicit trails over green areas within the town. These concerns were related to local knowledge and individual user perceptions that are only possible to integrate through descriptive data (if available) and personalization to the user's preference.

Our participants faced issues due to incorrect or incomplete data labelling in OSM, where the tours included paths that did not exist (3/31 tours) or crossed private properties (5/31). A path (1/31) was reported to have a very steep section. A possible reason can be that the steepness penalty factor used by the algorithm was too low. There were also closed paths (1/31) and barriers (2/31) due to fixed opening hours or constructions. While they had a limited impact overall, data inaccuracy and temporal disruptions could be solved by leveraging community-generated data to improve the initial tour recommendations. Other qualitative data concerned the navigation support included in our prototype. These are described in the Limitations section below since they are not directly related to the evaluated hypotheses but may have indirectly impacted the results.

5 DISCUSSION

5.1 Multi-data approach

We wanted to understand if map data alone was enough to generate pleasant itineraries or if runners' traces, as in Strava, were also needed through this evaluation campaign. Our approach requires map data to be available, accurate and pre-processed. Our algorithm relies on OSM data, where an impressive community edits streets, path, barriers, etc. While we faced many accuracy issues in the map data, we find that our algorithm indeed generates pleasant routes. However, a more personalized approach is required to recommend these routes to the user. Our participants expressed the desire to control both the algorithm and the interface aspects of the recommendation system. As in multi-criteria recommendation [1, 6], the user's individual preferences for specific item properties, or path characteristics, and the context data are necessary. While the location is important for recommending an appropriate route, attributes like ambience or safety and time of the day also need to be considered to satisfy the runner in the end. Hence, we conclude that a complementary approach of mining map data and combining contextual knowledge with explicit user preferences is needed for robust itinerary recommendation systems.

5.2 Itinerary Flexibility

Another approach to solving temporal disturbances and insufficient data would be to add flexibility in the itineraries and let users be free to modify them as they go. By supporting a dynamic adjustment of the proposed itinerary, we can address both unplanned factors (e.g. construction work, the weather, tiredness, etc.) and the subjective preferences of every runner, helping them get the most pleasant run. To support such flexibility, we can provide a set of alternative paths that may be interesting to explore for the runner [5] with each computed tour. The idea is to highlight alternatives to subsections of the original tour, roughly equivalent, in terms of pleasantness. These alternatives make it possible for the runner to deviate and explore beyond the precise initial itinerary, while still ensuring the quality and the match with their preferences.

6 LIMITATIONS

Our study had several limitations. One limitation is that in a real-world usability test as ours, it is difficult to eliminate the effect of confounding variables like weather conditions. Another limitation is that we had fewer participants than originally planned due to constant coronavirus COVID-19 pandemic restrictions and lockdown. It is a drawback for forming reliable statistical data. Other issues included improper or unsatisfactory navigation. 2/11 participants suggested the addition of voice on the app, 5/11 said that using a smartphone was impractical, especially in the mountains (1/11) for trail running and 2/11 participants said a watch would have been a more practical solution. The mobile prototype also had stability issues, and some (4/11) participants had to restart and try the tour again later. A supervised user test would have been able to tackle this situation by providing help on the spot without redoing the experiment. Also, a few participants could not rate the segments during the run (5 tours) due to technical glitches but rated them offline right after.

Despite the challenging circumstances, we were able to complete an initial study to validate our algorithm. It uncovered the potential issues and areas for improvement, contributing to research activities in this domain.

7 CONCLUSION

We developed an algorithm that constructs running tours that match the user's targets of distance and elevation gain, while maximizing the pleasantness potential. This algorithm's goal was to generate these tours without the need to rely on user-generated running traces. We defined the pleasantness characteristics of a path as proximity to nature, type of path (pedestrian, etc.), distance to high traffic roads, and path steepness. We mined that information from available map data. We conducted a user study with 11 participants to validate the algorithm's scores for these characteristics. The results show that the scores given by the algorithm had a positive correlation with the users ranking. However, we acknowledge the negative impact of missing data (safety, accessibility etc.) and wrong or obsolete data. Our solution also does not fully account for the complexity and variety of runners' personal preferences. To make our algorithm robust, we still need to tackle incorrectly labelled data issues, include contextual information, and allow personalization of the importance of different pleasantness characteristics. Two approaches of combining multiple data types and providing flexibility through alternatives in the running itinerary are discussed as possible solutions. Overall, the study showed our approach's potential to address the cold start problem in the generation of pleasant running tours in unknown locations.

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